

MATH5320 – Homework VIII

1. Credit risk mitigation

Consider the following three credit risk mitigants:

1. Netting
2. Collateralization
3. Use of a centralized clearing party

Explain what each is, how it reduces credit risk and describe one way that each one increases risk.

Credit risk mitigation refers to mechanisms to ameliorate the exposure of financial institutions to counterparty default. Three common approaches used are netting, collateralisation and usage of a central clearing counterparty.

Netting refers to the idea that we can reduce credit exposure by aggregating multiple bilateral transactions with the same counterparty into a single net obligation, such that instead of each contract being settled individually, where gains and losses across contracts are offset. This reduces the total exposure as the net m2m amount is a risk if the counterparty defaults. In practice, such netting agreements are implemented and regulated via ISDA master agreements. This is because, this can come from swaps, and if we are on the OTM leg, that is a liability, and is counted against us, whereas we are ITM, and the counterparty defaults, we have made a loss. The idea is that by netting, we can narrow such incidences.

As a result, it lowers current exposures, expected exposures, expected losses and CVA. However, netting introduces legal risk, since the enforceability of the netting agreement must be recognised under the relevant jurisdiction at the time of default, especially since many agreements are cross-border, and is subject to local regulations. It needs to be also noted that netting effectively changes the seniority structure of debt, and in that sense, is a transfer of risk from counterparty to bondholders.

Collateralisation, as its names suggest is obvious, is under the ISDA Credit Support Annex, which reduces credit risk by requiring counterparty with a negative mark-to-market position (or not having enough maintenance margin etc.) to post collateral, subject to a minimum transfer amount. This collateral serves as protection against losses if the counterparty defaults. Collateral is typically exchanged through variation margin, which adjusts daily according to market movements, and sometimes initial margin, which protects against potential future exposures.

Collateralisation, like the above, can reduce exposures. But it does introduce liquidity risk, since the counterparties must meet the margin calls immediately. Operational risks will also persist, since it is complex. Gap risk, can arise because exposures naturally can still change between margin calls.

Notably, collateralisation does not reduce risk to 0, and exposure is still equal to the thresholds plus movement in the market value over the period, and potentially the MTA itself,

and there is still residual exposure. The collateral may also be insufficient with huge volatilities.

The use of clearing houses, as third parties that stand between the original transacting parties, was mandated after the GFC. After clearing, each original party faces the CCP rather than each other. The goal is to improve the market, by having greater transparency, standardization, and the ability of the CCP to apply risk mitigation tools centrally, rather than having all the derivatives traded OTC.

The idea is the note that the risk is not eliminated, and while trades are balanced, so that the CCP has no ordinary market risk before default, it still bears the default risk. To manage this, CCPs themselves, have collateralisation, netting, default settlement rules and clearing house capital. In the cause of defaults, there are management tools such as auctioning off trades, reassigning trades, variation margin gains haircut, and tear-ups.

As such, the use of CCPs reduces bilateral counterparty risk, but centralises the risk. IF the margins and other protections are insufficient, the losses can arise the clearing house level.

Cumulatively, all three mitigants reduce counterparty risk, but none of them makes risk disappear, each with their own idiosyncrasies. It *mitigates* but does not eliminate risks whatsoever.

2. CDS

Assuming a constant hazard rate of 0.03, a risk free rate of 5%, a 40% recovery rate, and annual coupon payments, what is the par spread for a CDS contract maturing in 5 years? In 10 years?

For the above given parameters:

```
import pandas as pd

# Given parameters
r = 0.05          # risk-free rate
lam = 0.03        # hazard rate
R = 0.40          # recovery rate
```

CDS contract pricing mainly comes from the value of three cash flows:

CDS contract pricing.

- Default time τ .
- Pick an equivalent martingale measure Q with respect to a numéraire N .

Valuation of cashflows:

- Value of each coupon: $N_0 E^Q [C \alpha(t_i) 1_{\tau > t_i} / N_{t_i}]$
- Value of accrued interest: $N_0 E^Q [C \alpha^*(\tau) 1_{\tau \leq t_n} / N_\tau]$
- Value of floating leg payment: $N_0 E^Q [(1 - R) 1_{\tau \leq t_n} / N_\tau]$

We intend to price the CDS in accordance to what was given in the lecture such that (derived through RN-Pricing with the assuming of independence of default time and the numéraire, where R is also a known constant):

- $D(t) = e^{-rt}$ be the discount factor,
- $s(t) = P(\tau > t)$ be the survival probability,
- $p(t)$ be the default density.

Under a constant hazard-rate reduced-form model (not realistic):

$$s(t) = e^{-\lambda t}, p(t) = -\frac{d}{dt} s(t) = \lambda e^{-\lambda t}$$

For a CDS with maturity t_n , the par spread formula is given as follows:

$$C(t_n) = \frac{\int_0^{t_n} (1 - R) D(u) p(u) du}{\sum_{i=1}^n a(t_i) D(t_i) s(t_i) + \int_0^{t_n} a^*(u) D(u) p(u) du}.$$

Notably, here the numerator is the “protection” leg, wherein the denominator is the “premium leg”, which consists of scheduled coupon payments and the accrued premium paid at default. Notably, the accrual factor is 1, as it is stated we have annual coupon payments.

We yield the following results:

Maturity	Protection Leg PV	Premium Leg PV	Fair CDS Spread	Fair CDS Spread (bps)	
0	5	0.074178	4.019348	0.018455	184.552297
1	10	0.123901	6.713597	0.018455	184.552297


```
CDS maturity = 5 years
Protection leg PV = 0.07417799
Premium leg PV   = 4.01934796
Fair CDS spread  = 0.01845523
Fair CDS spread  = 184.55 bps
-----
CDS maturity = 10 years
Protection leg PV = 0.12390098
Premium leg PV   = 6.71359747
Fair CDS spread  = 0.01845523
Fair CDS spread  = 184.55 bps
```

Notably, here the fair 10y and 5y oar CDS spread are the same in this model, because in this case, we assume a constant hazard, recovery and risk-free rate. And under these assumptions,

the term structure of default intensity is flat, so the CDS spread curve is also flat. Intuitively, both the protection leg and premium leg scale consistently with maturity under the exponential survival, so the fair spread does not change with maturity.

We note:

▲ The chapter in [Hull](#) on Credit Risk gives the same formula as emcor as a first approximation with a justification:

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Consider first an approximate calculation. Suppose that a bond yields 200 basis points more than a similar risk-free bond and that the expected recovery rate in the event of a default is 40%. The holder of a corporate bond must be expecting to lose 200 basis points (or 2% per year) from defaults. Given the recovery rate of 40%, this leads to an estimate of the probability of a default per year conditional on no earlier default of $0.02/(1 - 0.4)$, or 3.33%. In general

$$\bar{\lambda} = \frac{s}{1 - R}$$

where $\bar{\lambda}$ is the average default intensity (hazard rate) per year, s is the spread of the corporate bond yield over the risk-free rate, and R is the expected recovery rate.

This formula can easily be rewritten as

$$s = \bar{\lambda}(1 - R)$$

as pointed out by @emcor.

In our above case, we note that:

$$(1 - R)\lambda = (1 - 0.40)(0.03) = 0.018,$$

so

$$0.018 \times 10,000 = 180 \text{ bps.}$$

So results do match up, such that the PV of fixed leg = PV of the protection leg (floating). This is the case under a constant hazard rate model.

Also, we note that the protection leg and premium leg present values, as the CDS spread is chosen to make:

$$C \times (\text{premium-leg annuity}) = \text{protection leg} \rightarrow C = \frac{\text{protection leg}}{\text{premium-leg annuity}}$$

Numerically, it should be noted that,

$$C = \frac{0.07417799}{4.01934796} \approx 0.0184$$

Which makes sense, as the fair spread. The 10Y CDS covers a far longer time period, so both of the legs accumulate more present value, and recalling the formulas:

Protection leg:

$$\int_0^T (1 - R)D(u)p(u) du$$

Premium-leg annuity:

$$\sum_{i=1}^n a(t_i)D(t_i)s(t_i) + \int_0^T a^*(u)D(u)p(u) du.$$

We note the T that are within the integrals that are the weighted probability of defaults, discounted by time etc. and in this case the protection leg includes defaults over Y6-Y10, so it is larger, the premium leg part includes more coupon dates, and more accrued premiums. So, intuitively, the numbers do make sense.

3. Monte Carlo VaR I

Same setting as in previous VaR calculation problems.

Compute the 99% 5 day VaR and 97.5% 5 day ES of P by Monte Carlo. Simulate it based on the volatility estimates for P from previous problems using the parameter estimates based on 5 year windows and assuming that P follows GBM. Again, compute it each day assuming a \$10,000 position in P each day.

Graph the results and compare to the relevant previous versions of VaR calculations that used 5 year windows.

Which VaR calculations does the Monte Carlo VaR agree with, and which does it differ from? Why?

Recalling the previous homework setup (Homework VII):

Let S being a portfolio P consisting of about \$5,000 invested in AAPL and about \$5,000 invested in CAT, where the investment was made about 20 years ago, say on 1997-10-13. On that date, the price AAPL was 0.2026 and the price of CAT was 29.3125, so that would yield 24679 shares of AAPL, and 171 shares of CAT.

Homework VI:

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Assuming S is GBM, compute the portfolio mean and variance with 2, 5, and 10 year windows, and with the equivalent exponential weighting λ s of 0.9972531953, 0.9989003714, and 0.9994500345, respectively. Compare the window based estimates of the mean and variance with the corresponding exponentially weighted ones over the history and graph the results.

Here, the 95% 50day VaR and 97.5% 5-day ES for the portfolio by MC, using the same setup as the previous problem, with a scaled \$10,000 portfolio for our VaR and ES. We use a rolling 5-year window for this, and assume that the portfolio itself follows GBM, and compute the risk measures for each day, as shown below with the head and tail of the time series:

	Dates	Q3_MC_VaR95_5d	Q3_MC_ES97.5_5d
0	1987-11-09	983.199248	1374.294811
1	1987-11-10	954.518228	1387.599432
2	1987-11-11	960.882753	1346.898329
3	1987-11-12	979.773918	1379.201776
4	1987-11-13	980.416232	1380.616746

	Dates	Q3_MC_VaR95_5d	Q3_MC_ES97.5_5d
9632	2026-02-05	589.938639	841.054399
9633	2026-02-06	581.513829	816.690539
9634	2026-02-09	562.586854	798.134162
9635	2026-02-10	595.750054	849.734694
9636	2026-02-11	582.219668	818.933088

Following this, we set up the portfolio, such that it starts off initially with \$5,000 invested in AAPL, and \$5,000 invested in CAT on 1997-10-13. The idea is that from this, we estimate parameters with a rolling window, rescale to a fixed dollar position on that day, and under GBM assumption, we work with the mean and variance of log returns.

Notably, we also compare to previous exponentially weighted estimators (with the same param given in the solutions (as shown above)).

We modelled in the way such that we have P_t denote the value of the portfolio, where under the GBM assumption we have:

$$\frac{dP_t}{P_t} = \mu_P dt + \sigma_P dW_t$$

And over the horizon, we have a normally distribution of continuously compounded (we assume they are iid), such that we have additive returns, wherein we have:

$$r_t^P = \log\left(\frac{P_t}{P_{t-1}}\right), \text{ which is estimated using the trailing 5 year window,}$$

And then, using the trailing 5-year window, and then estimate:

$$\hat{\mu} = \text{sample mean of } r_t^P, \quad \hat{\sigma} = \text{sample standard deviation of } r_t^P.$$

For a 5-day horizon, we use the scaling:

$$\mu_5 = 5\hat{\mu}, \quad \sigma_5 = \sqrt{5} \hat{\sigma}.$$

Noting that the scaling is standard for iid, normally distributed returns. We then simulate the returns :

$$R_5^{(j)} \sim N(\mu_5, \sigma_5^2), \quad j = 1, \dots, N,$$

But the simulated future value of a \$10,000 position is such that:

$$V_{t+5}^{(j)} = 10,000 e^{R_5^{(j)}}.$$

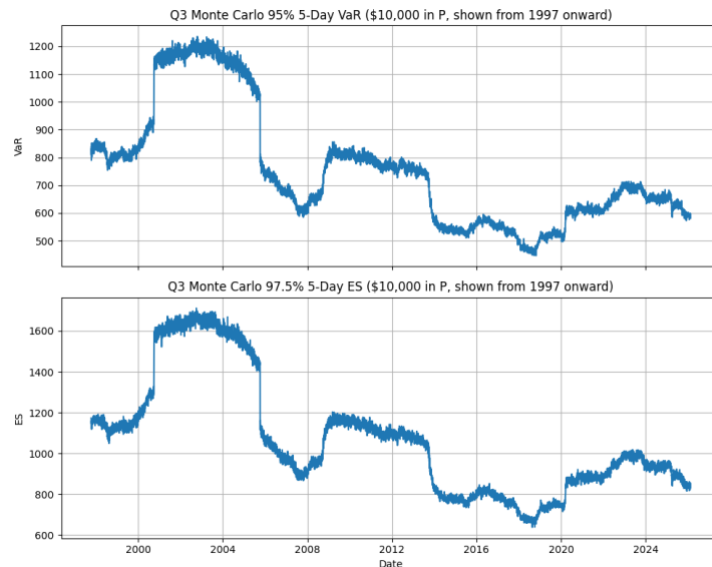
Then from this, we have the following simulated loss of:

$$L^{(j)} = 10,000 - V_{t+5}^{(j)}.$$

Then, by convention, we take the 95th percentile of losses for the VaR, and the expectation or the average of the losses beyond the 97.5 percentile. For the most recent date of data, we have:

- 95% 5-day VaR = 593
- 97.5% 5-day ES = 850

That is expected, with ES substantively above the Var, with time variation in risk, as shown below:



Comparing with other methods, with the historical simulation, we can see that the Monte Carlo VaR agrees most closely with the GBM/lognormal parametric method. This is largely expected, because both methods are built on the same assumptions of the portfolio assumed to follow GBM, the mean-var are estimated from a rolling 5y window, and the losses are from the lognormal return structure. Consider the following results, where we use the most recently available data.

	Model	95% 5-day VaR	97.5% 5-day ES
0	Historical Simulation	561.009040	958.137588
1	GBM/Lognormal Parametric	586.748917	853.450191
2	EWMA ($\lambda=0.9989003714$)	317.509463	586.631504
3	Q3 Monte Carlo (portfolio GBM)	593.521566	850.981762

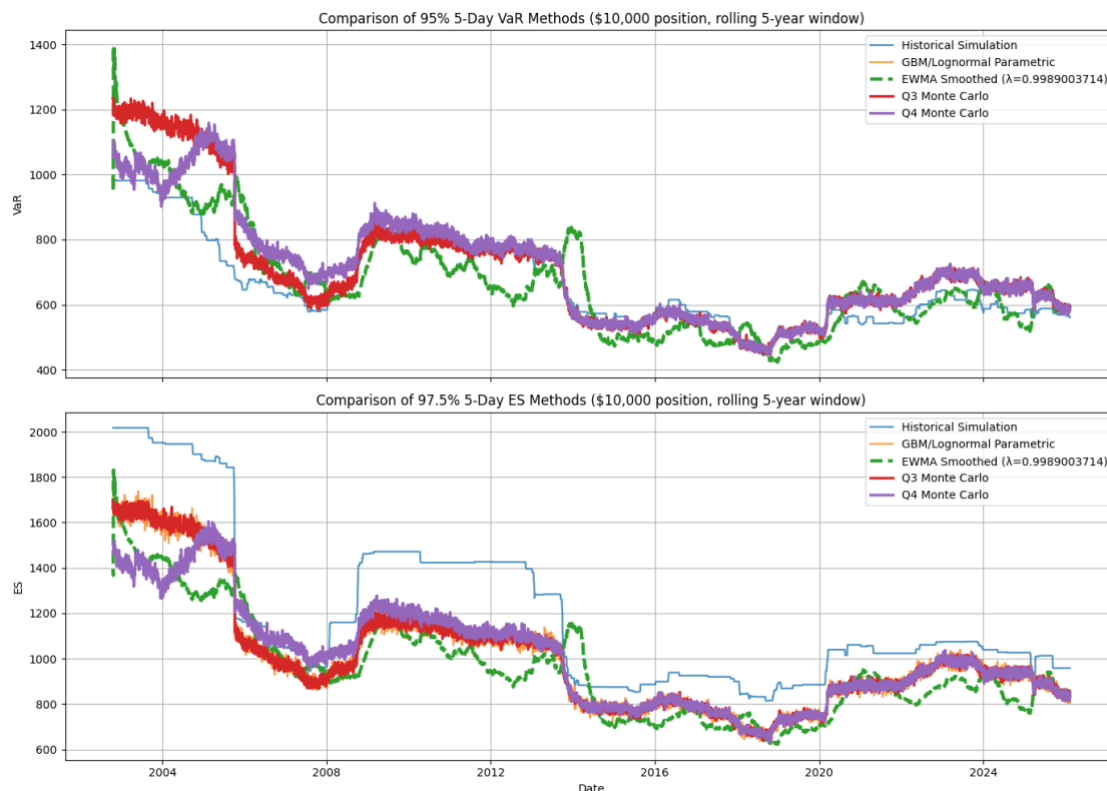
Notably, the Q3 Monte Carlo method differs more from historical simulation, especially in ES. The VaR remains reasonably close, but the historical ES is larger. This is consistent, as historical simulation preserves the realised tail events (usually fatter in reality) in the trailing 5-year window, including skewness, jumps, and unusually large losses. By contrast, the Monte Carlo method imposes the GBM/lognormal structure (which we know isn't true in reality), which smooths the loss distribution and typically produces thinner tails than the empirical historical distribution. Therefore, historical simulation can generate a larger ES if the sample window contains particularly severe losses

	Model	95% 5-day VaR	97.5% 5-day ES
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Noting the EWMA, it differs the most. EWMA places greater weight on more recent observations, and much less weight on older ones. Since the 5-year equivalent exponential-weighting parameter is very close to 1, the estimator is still persistent, but it remains more sensitive to recent volatility conditions than the plain rolling-window estimator. The difference suggests that recent volatilities are more calmer than the broader 5y history reflected in the rolling window MC simulation.

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As shown on a historical basis also, the MC VaR closely tracks the GBM method, as they rely on the same assumptions. They differ on the fact that the parametric model computes the quantiles using the normal distribution, whereas the MC method approximates the same distribution through simulation. Because the underlying distributional assumptions are identical, the resulting VaR and ES estimates are very similar.



To make the differences clearer, the following figure shows a zoomed-in comparison of the methods over recent years. The zoomed view highlights that the Monte Carlo and parametric models remain closely aligned, while the EWMA estimator fluctuates more in response to short-term volatility changes. This occurs because both methods assume the same GBM/lognormal return structure, with Monte Carlo approximating the same distribution numerically rather than computing quantiles analytically. Historical simulation remains somewhat more irregular due to its reliance on realised return observations rather than a parametric distribution.



On an aggregate basis, this Q3 Monte Carlo VaR agrees with the GBM or parametric method, as both methods assume that P follows GBM, and use the same 5y rolling idea. It differs from the historical simulation, especially for ES, since it preserves the realised tail events, rather than imposing a smooth GBM/lognormal distribution. It also differs from exponential weighting, as it places more emphasis on recent observations, and therefore gives materially lower risk estimates when in a low vol period.

4. Monte Carlo VaR II

Repeat the last problem, but this time compute the VaR by simulating the underlying stocks, which are assumed to follow GBM driven by correlated Brownian motions, and estimated using 5 year windows. You will need to work out how to estimate the correlation.

Graph the results and compare to the appropriate previous VaR calculations using 5 year windows.

Which VaR calculations does the Monte Carlo VaR agree with, and which does it differ from? Why?

We repeat the previous Monte Carlo VaR exercise, but instead of modelling the portfolio P directly as a GBM, we model the two underlying stocks, AAPL and CAT, as correlated geometric Brownian motions. Setup remains the same as previously with a \$5,000 allocation to both stocks for the portfolio. The simulation is done on the stock level, rather than on the portfolio level.

We assume that the two stock prices follow correlated GBMs:

$$\frac{dS_t^{AAPL}}{S_t^{AAPL}} = \mu_{AAPL} dt + \sigma_{AAPL} dW_t^{AAPL},$$
$$\frac{dS_t^{CAT}}{S_t^{CAT}} = \mu_{CAT} dt + \sigma_{CAT} dW_t^{CAT},$$

With the following:

$$dW_t^{AAPL} dW_t^{CAT} = \rho dt.$$

This is literally from STAT5264, where the dt term survives as a result of the quadratic variation.

And then, we note that the two BMs are correlated with the Brownian motions are correlated via ρ and as such, in discrete timing, this means that the vector of daily log returns is modelled as a bivariate normal, wherein:

$$\Sigma = \begin{pmatrix} \sigma_{AAPL}^2 & \rho \sigma_{AAPL} \sigma_{CAT} \\ \rho \sigma_{AAPL} \sigma_{CAT} & \sigma_{CAT}^2 \end{pmatrix}$$

The correlation parameter ρ is estimated from the sample correlation of AAPL and CAT log returns over the rolling 5-year window. In practice, this is obtained from the empirical covariance matrix of the two return series:

$$\Sigma = \begin{pmatrix} \sigma_{AAPL}^2 & \rho \sigma_{AAPL} \sigma_{CAT} \\ \rho \sigma_{AAPL} \sigma_{CAT} & \sigma_{CAT}^2 \end{pmatrix}$$

where the entries are computed directly from the sample moments of the historical returns in the rolling estimation window.

Intuitively, this means that the mean vector, volatilities, and correlation are all estimated using the trailing 5-year window of historical returns. In particular, the correlation is obtained from the sample covariance matrix of AAPL and CAT log returns over the rolling window.

Then, exactly like before, we scale like the following, but with the covar matrix such that:

$$\mu_5 = 5\mu, \quad \Sigma_5 = 5\Sigma,$$

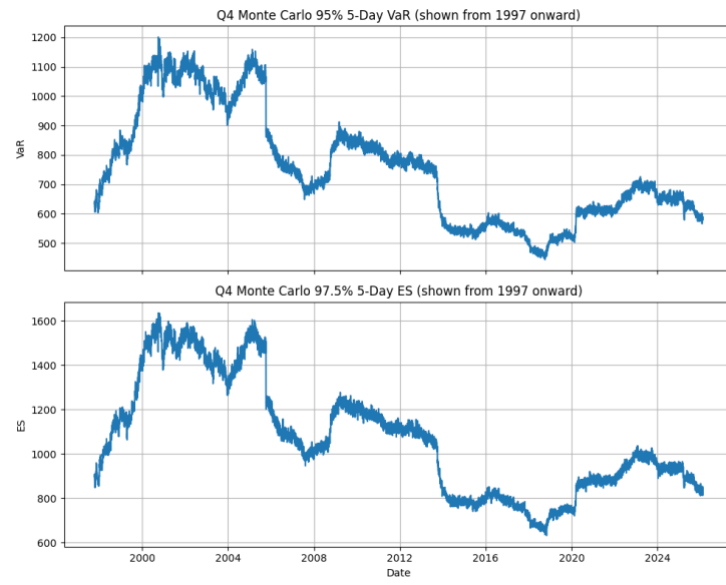
And we simulate the following:

$$R_5^{(j)} \sim N(\mu_5, \Sigma_5), \quad j = 1, \dots, N$$

Then we have simulated stock prices which are like:

$$S_{t+5}^{(j)} = S_t e^{R_5^{(j)}},$$

And as such, from here the simulated portfolio is constructed from the simulated stock prices, which are then compressed to \$10,000 notional, and the rest of the simulation is the same as before. Like before, it is notably time-variant and dependent on the regimes:



The resulting rolling Q4 VaR and ES series begin at the portfolio inception date and evolve through time using rolling 5-year estimates. For the most recent date in the sample, the Q4 Monte Carlo results are approximately:

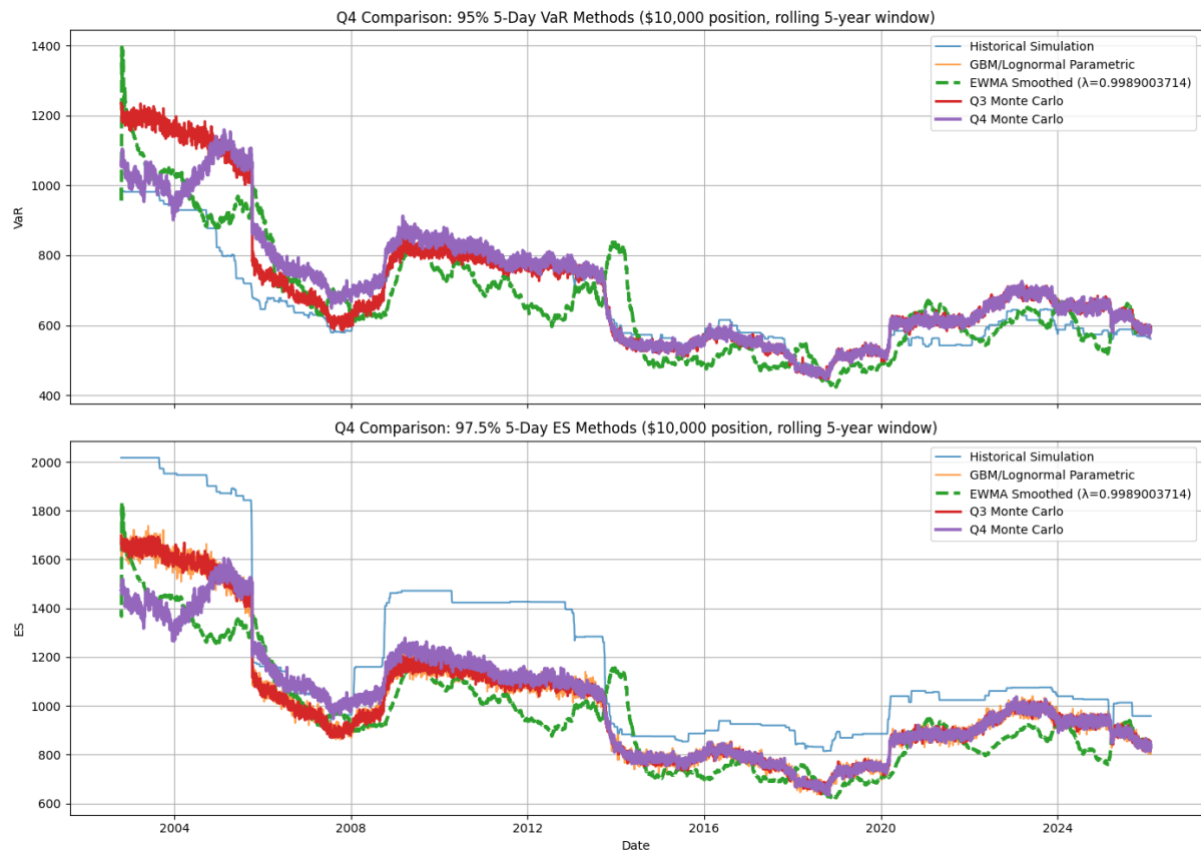
- 95% 5-day VaR = 587
- 97.5% 5-day ES = 829

Notably, it does appear to be lower than historical values.

Now moving onto the comparison with the other methods. We first compare on the final day available in the data.

	Model	95% 5-day VaR	97.5% 5-day ES
0	Historical Simulation	561.009040	958.137588
1	GBM/Lognormal Parametric	586.748917	853.450191
2	EWMA ($\lambda=0.9989003714$)	317.509463	586.631504
3	Q3 Monte Carlo (portfolio GBM)	593.521566	850.981762
4	Q4 Monte Carlo (underlying correlated GBMs)	586.715880	828.905182

It is the most meaningfully similar to the previous question of Monte Carlo and the GBM/lognormal parametric method. This is expected. Q3 simulates the portfolio P directly under a GBM assumption, whereas Q4 simulates the two underlying stocks and then reconstructs the portfolio. This is because they follow the identical core assumptions of lognormal prices, normal log returns, and rolling 5y parameter estimations. The only difference is that the portfolio distribution is built from the joint simulation of the two stocks directly, rather than the portfolio directly.



Notably, it does remain fairly similar to the GBM/lognormal parametric VaR, because that method is still based on the same normal/lognormal structure. The Monte Carlo method simply computes the distribution numerically rather than analytically.

Exactly like before, the current pre-stock MC differs more from historical simulation, especially in ES. Historical simulation preserves realized tail events and any empirical non-normality in the 5-year window, including skewness and unusually large negative returns. As a result, it can produce higher tail-risk estimates than model-based Monte Carlo methods.

Same idea for EWMA. Same for reasoning.

The findings are more pronounced, as shown here for the last few years, where like mentioned, the MC simulations are the most similar to the GBM/lognormal VaR and ES, where it differs the most with historical.



Since Q4 simulates the underlying stocks jointly rather than the portfolio directly, it provides a useful check on whether modelling the portfolio as a single GBM in Q3 is a good approximation. The close agreement between Q3 and Q4 suggests that, in this case, the portfolio-level GBM approximation is reasonably accurate.